

Data Preprocessing & Cleaning Report

Project: Out-of-Pocket Healthcare Costs and Poverty Study in Kolhapur City

1. Dataset Overview

The primary data was gathered using a structured bilingual (English/Marathi) survey questionnaire distributed to respondents across Kolhapur city. The raw input contains **300 patient observations** with **29 distinctive response columns** recording detailed demographic characteristics, health system selections, custom cost categories, financial constraints, and insurance details.

Metric Property	Value Details / Findings
Raw Dimension	300 Rows × 29 Columns
Target Variables	Demographics (Age, Gender, Income), Specific Expenditures (Consultation, Medicine, Travel, Diagnostics), Financial Strain Metrics
Primary Challenges	Bilingual text mix, uncleaned encoding characters (e.g., â€”), categorical range responses requiring numeric conversion, and structured null value imputations.

2. Sequential Data Cleaning Pipeline

The raw dataset was cleaned and transformed using **Python (Pandas, Re, and NumPy)** through the following programmatic phases:

Step 1	Feature Pruning & Row Initialization The initial column Timestamp, generated automatically by the survey intake tool, was dropped from the dataset as it holds no analytical weight for statistical correlation. <pre>df = df.iloc[:, 1:]</pre>
Step 2	Bilingual Column Rename Mapping Long, verbose column headings written in combined English and Marathi were shortened into explicit, structured Snake_Case variables to allow trouble-free coding execution. <pre>name = ['Age_group', 'Gender', 'Household_size', 'Family_type', 'Residence_area', ...] df.columns = name</pre>
Step 3	Encoding Defect Correction & Character Sanitization

Special symbols and bad text encoding breaks (e.g., â€” or dashes) were systematically targeted using regular expressions to isolate the standard clean English categories.

```
def replace_dash(x):  
    return str(x).replace("-", "/").replace("â€”", "/")
```

Step 4

Bilingual Text Stripping & Regular Expression Filtering

A custom string parsing function was written to drop the trailing Marathi text explanations (split by "/") and strip text inside brackets to leave pure categorical factors.

```
def clean(x):  
    x = str(x).split("/")[0]  
    x = re.sub(r"\s*(\.*?\)", "", x)  
    return x.strip()
```

Step 5

Logical Cross-Variable Inferences

To fix structural survey consistency, responses representing public health treatment routes were evaluated. For patients who chose a Government hospital, operational costs like hospital paper costs, diagnostics, and consultation fees were logically zeroed out.

```
df.loc[df['Type_of_hospital'] == 'Government hospital', ['Hospital_visit_cost',  
'Consultation_fee', 'Diagnostics_cost']] = 0
```

Step 6

Categorical Interval to Numeric Midpoint Mapping

To enable mathematical analysis, textual expense intervals (such as 0-500, 101-500) and financial intervals were decoded. Multi-word string structures were split, extracting boundary numeric digits to map text records directly into continuous midpoints or upper numeric caps.

```
if "-" in x:  
    parts = x.split("-")  
    return (float(parts[0]) + float(parts[1])) / 2
```

Step 7

Anomalous Outlier Truncation & Null Treatment

Rows containing non-standard demographic records (such as Gender: Other) were filtered out to avoid thin sample biases. Missing value counts across the target series 3_month_total_expenditure were safely imputed using the calculated series mean, and missing scale factors were filled with column modes.

```
df['3_month_total_expenditure'] =  
df['3_month_total_expenditure'].fillna(df['3_month_total_expenditure'].mean())
```

Step 8

Dataset Cohesion & Feature Export

The isolated string categories and transformed numeric columns were stitched back together into a final processed matrix, rounded off cleanly, and exported as Hospital Data.csv for descriptive statistics and modeling stages.

```
final_df = pd.concat([df_selected, new_df], axis=1)
final_df.to_csv("Hospital Data.csv", index=False)
```

Report Conclusion: The preprocessing pipeline successfully converted messy, bilingual raw questionnaire logs into a neat tabular format. Outliers, text ranges, and structural anomalies are successfully fixed. The output dataset is fully optimized for modeling healthcare expenditures and calculating poverty impacts in Kolhapur.